
Breakout

Analysis

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Table of Contents

Table of Contents	1
1 Introduction	2
2 Project description	Error! Bookmark not defined.
2.1 Problem and intended users	Error! Bookmark not defined.
2.2 Personal learning goals	Error! Bookmark not defined.
3 Scope and constraints	Error! Bookmark not defined.
4 Analysis approach and findings	2
4.1 Research methods to use next	2
4.2 Scope decisions	2
5 Requirement priorities	2
6 Must have requirements	3
7 Risks and validation priorities	Error! Bookmark not defined.
8 Sources	Error! Bookmark not defined.

1 Introduction

This analysis defines the problem, scope and requirements for my Breakout project. My aim is to build a small, playable Windows desktop game while developing my Java and object-oriented programming skills. The main priorities are reliable gameplay, clear feedback and a scope that leaves enough time for testing. Breakout Advice explains the recommended solution and planning.

2 Analysis approach and findings

The workshop describes analysis as a structured investigation of the context, goals, stakeholders and constraints. For this project, reviewing the game design and initial documents establishes the intended behaviour. A literature review of framework documentation supports the technology comparison in Breakout Advice. These activities do not establish that the game has already passed testing.

Question	Current finding	Further evidence needed
What must the player achieve?	Clear every brick before losing three lives.	Observe whether a new player understands the goal from the start screen.
What is the minimum viable product?	One complete session with movement, collisions, score, lives and restart.	Demonstrate both endings and a clean restart.
Which technology fits the learning goals?	Existing Java knowledge favors a Java solution.	Compare framework support and confirm a working Windows prototype.
What creates the greatest technical risk?	Collision response and state resets can make an otherwise simple game unreliable.	Test edge hits, high speeds, life loss and repeated restarts.

2.1 Research methods to use next

Available product analysis can compare how existing brick-breaker games explain controls, display lives and handle restart. I will record the products inspected and observations before using them as evidence. An expert interview with a teacher can check the scope and collision approach. Short peer playtests can check whether the intended audience understands the controls and feedback. These are planned activities, not completed research results.

2.2 Scope decisions

Windows and a single level are defined as the baseline. Extra platforms and multiple layouts are deferred. The earlier Analysis proposes a paddle movement “flick” effect, but the current design only describes changing the bounce angle by hit position. Hit-position control is therefore a Should have, while the flick effect is excluded from the first version.

Points per brick and a final ball speed are not yet specified. Before validation, I will record the chosen fixed points value and tune the supported speed through playtesting. The Testing and Risk Analysis notes currently repeat requirements. The acceptance checks below provide a basis for a separate test record with actual results.

3 Requirement priorities

Must have requirements define a complete playable version. Should have requirements improve the experience after that core works. Could have features depend on remaining time. Won't have features are deliberately excluded from this version. The identifiers below replace the earlier M01 to M10 list and retain the original identifiers REQ-01 to REQ-11 for traceability.

4 Must have requirements

Each acceptance check describes how to verify the finished game. The checks are planned and do not imply a passing result.

ID	Requirement	Acceptance check
REQ-01	Run as a desktop application on Windows.	Launch using the documented setup instructions on the target Windows computer without a blocking error.
REQ-02	Move the paddle left and right	Test each key separately and confirm the matching direction.
REQ-03	Keep the whole paddle inside the play area.	Hold each movement key at the corresponding edge. No part of the paddle leaves the area.
REQ-04	Move the ball automatically during active play.	Start a session and release all controls. The ball continues moving.
REQ-05	Bounce from the top and side walls, paddle and bricks.	Test each surface, including brick side and corner hits. The ball leaves the contact without sticking or repeatedly reversing there.
REQ-06	Remove a brick when the ball hits it.	Hit a brick and confirm it disappears and cannot score again.
REQ-07	Display and update the score during gameplay.	Start at zero. Destroy three bricks and confirm the total equals three times the configured points per brick.
REQ-08	Display starting lives and deduct one per missed ball.	Miss once. Lives change from three to two exactly once and the ball resets. Repeat until zero.
REQ-09	Show the win screen when all bricks are destroyed.	Remove the final brick. Active play stops and the win screen shows the final score.
REQ-10	Show game over when no lives remain.	Miss the ball with one life remaining. Active play stops and game over shows the final score.
REQ-11	Restart from either ending with R. design	Press R after a win and a loss. Bricks return, score becomes zero, lives become three and a new session can be played.
REQ-12	Show the title, controls and Space start prompt.	On launch, the start screen appears. Space begins gameplay.